

Image Processing and Pattern Recognition

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Instructions for dear students,

- Please submit your assignment before the designated submission date. Late submissions will not be considered for grading.
- Create separate answer sheets for this assignment. Remember to collect your submitted sheets once the assignment has been checked.
- This assignment contributes a total of 10 marks to the internal mark evaluation criteria outlined by Pokhara University.
- If you encounter any issues, feel free reach out at harimohan.khatri@eemc.edu.np or visit in person.

CHAPTER – 01

1. Define digital image, digital image processing and explain fundamental steps of digital image processing.
2. Explain elements of digital image processing systems.
3. Define sampling and quantization. Explain the process of image acquisition in detail.
4. Differentiate between spatial resolution and intensity resolution of an image? What are the effects of both resolutions on image enhancement process? Explain.
5. How adjacency of the pixels affects connectivity and path between the pixels? Explain with suitable examples.
6. Define City-block, Chessboard and Euclidean distance.

CHAPTER – 02

1. Define point operation. How can you determine the negative of an image? Mention applications of digital negative.
2. Perform contrast stretching in given image data.

a)

r	0	1	2	3	4	5	6	7
n	0	21	35	0	44	88	7	0

b)

2	3	1	4
7	5	6	6
3	6	5	5
3	2	2	5

3. What do you mean by gamma correction? Differentiate between log transformation and power transformation.
4. Define Thresholding. Differentiate between intensity level slicing and bit plane slicing.
5. How do we represent image statistically using histogram? Can we change image statistics according to any reference histogram data? If yes. Explain how?
6. Perform histogram equalization in given image data.

a)

r	0	1	2	3	4	5	6	7
n	0	21	35	0	44	88	7	0

b)

2	1	1	4
4	1	4	3
3	2	3	3
3	2	2	4

c) [5,5,3,3,3,4,4,2,1,1,1,1,0,0,4,3,2,0,0,5,4,3,1,1]

7. List out different arithmetic operations with necessary description in image processing. How arithmetic operations can be used for shading correction and masking in an image.
8. List out different set operations with necessary description in image processing. How digital negative of an image can be calculated using set operations? Explain.
9. Explain different logical operations in image processing. Give operation examples with binary image samples.
10. Define neighborhood operation in image. How it differs from point operation. Explain correlation and convolution process with necessary expressions.
11. How pixels around the edges are handles while performing convolution process? Explain each solution types.
12. Define padding. Perform zero, wrap around, mirror and replication padding in given image data for 5*5 kernel.
13. What are the limitations of box filter? Explain Gaussian low pass filter for image smoothing?
14. Perform average weightage filter in given 4*4 image.

2	1	1	4
4	1	4	3
3	2	3	3
3	2	2	4

15. Differentiate between linear and non-linear spatial filters. Explain different non-linear spatial filters for image smoothing. Also explain working of each filter for given image data.

200	100	105	104
40	11	14	33
39	24	39	73
63	82	82	14

16. How 1st difference and 2nd differences can be used to sharpen the image? Derive Laplacian filter kernel for enhanced image.
17. How high-boost filter works? Explain Robert's cross gradient, Prewitt's and Sobel's filter for image enhancement.
18. Explain different techniques used for magnification of an image with suitable examples.

CHAPTER – 03

1. Define Fourier Series, Fourier Transform and Discrete Fourier Transform.
2. Explain Properties of Discrete Fourier transform with their application in image processing.
3. When do we need to center the Fourier transform of an image? Prove that multiplying the image intensities by $(-1)^{x+y}$ where (x,y) represents coordinates of pixels, centers the Fourier Transform?
4. Write short notes on Fast Fourier Transform.
5. List out the steps involved in frequency domain enhancement of image.

6. Differentiate between Ideal, Butterworth and Gaussian smoothing filter used in frequency domain enhancement.
7. How high pass filters in frequency domain can be obtained from low pass filters? Explain different high pass filters obtained from low pass filters with their application areas.
8. Obtain Laplacian equivalent equation in frequency domain.
9. Derive kernel for Hadamard Transform with order $N=8$. Also mention its properties.
10. Explain natural ordering and sequential ordering in Hadamard transform. How sequential ordering can be obtained from natural ordering? Explain the techniques.
11. Calculate Forward and Inverse Hadamard transform for given 4×4 image.

4	3	3	4
5	1	5	3
3	6	3	9
2	3	1	4

12. Derive kernel for Haar Transform with order $N=2$. Also mention its properties.
13. Calculate Forward and Inverse Haar transform for given 4×4 image.

1	4	6	8
6	3	5	6
4	5	9	4
8	5	4	4

14. Define Discrete Cosine transform. With its properties and application areas. Perform forward and reverse DCT for given image data.

45	67	75	33
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CHAPTER – 04

1. Explain the model of image degradation and restoration with necessary block diagram.
2. Define White noise, Gaussian noise and Impulse noise. How impulse noise can be reduced using spatial filtering? Explain.
3. Write short notes on adaptive median filtering for noise reduction.
4. Define Periodic noise? Explain the process of reducing periodic noise using notch filtering.

CHAPTER – 05

1. Define redundancy in image data and its types.
2. Why do we need to compress the image? Differentiate between lossless and lossy compression. Explain generic model of compression for both types.
3. For given image data:

Code-1: Uses fixed length coding of 3 bit for each symbol.

Code-2: Uses variable length coding (Huffman Coding)

r	0	1	2	3	4	5	6	7
n	25	16	4	8	12	81	56	77

Calculate compression ratio and relative redundancy.

4. Define information theory for image compression. Calculate L_{avg} , Entropy and Efficiency for Huffman coding in given image data.

a)

<i>r</i>	0	1	2	3	4	5	6	7
<i>n</i>	0	21	35	0	44	88	7	0

b)

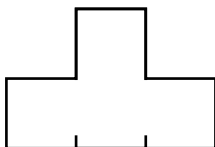
r	0	1	2	3	4	5	6	7
$P(r)$	0.01	0.02	0.04	0.22	0.4	0.1	0.03	0.18

5. Write short notes on:
 - a) Run Length Coding
 - b) LZW Coding
6. Explain predictive coding for lossless and lossy compression with necessary block diagram.
7. What are the parameters affecting the performance of transform coding? Explain transform coding process with necessary block diagram.
8. Compress the 4*4 image with transform coding with given parameters:
 - a. Transformation = Hadamard
 - b. Size of sub image = 4
 - c. Quantization = Divide by 10 and round off to nearest integer.

46	61	72	34
54	56	61	22
46	61	72	34
33	54	72	61

CHAPTER – 06

1. Define features. Differentiate between local and global features. Explain the role of feature detection and feature description in feature extraction process.
2. How image can be segmented using discontinuities in image? Explain each spatial filters to detect point, line and edges in image.
3. Write short note on Mexican Hat filter.
4. Explain edge linking for segmentation? Explain Hough transform for global edge linking.
5. Define thresholding. How do you determine good thresholding value for segmentation? What happens if we cannot determine good threshold value? How adaptive thresholding solves such issue? Explain with suitable example.
6. Explain Otsu's algorithm for optimum thresholding.
7. How edges can be used to enhance the segmentation process? Explain.
8. What do you mean by segmentation based on similarities? Differentiate between seeded and unseeded process.
9. Write short notes on:
 - a. Region growing based segmentation
 - b. Region splitting and merging
10. Differentiate between chain code and crack code with suitable example. How polygon approximation process can be used to describe the boundary region? Explain.
11. Define signature. Show that shape numbers are independent to change in starting point and rotation by 90° with the help of given shape.



12. How does the order of a descriptor determine the efficiency of shape representation? Explain the process of Fourier descriptor can be extracted?

CHAPTER – 07

1. What do you mean by image patterns? Explain types of patterns in image.
2. What are the general steps in image pattern recognition system? Explain each step with suitable block diagram.
3. Everest Engineering College, Sanepa intends to implement a face recognition-based automatic attendance system. As a system developer, outline the key steps involved in designing and developing such a pattern recognition system. For each step, specify the techniques or methods you would use and justify your choices with appropriate reasons.
4. Explain each step involved in SIFT feature extraction. How SIFT features can be used for image classification and decision making?
5. Briefly explain principal component analysis for image pattern recognition?
6. Define pattern and pattern class and pattern vectors. How image can be classified using prototype matching techniques. Explain each type.
7. Write short notes on
 - a. Bayes classifier
 - b. Perceptron
 - c. Hopfield Network